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LECTURE NOTES

# Discrete Mathematics notes

*Number Theory*

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DM549 - Discrete Mathematics

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# 1 Divisibility

## Definition: Divisibility

We say that a number  $a$  divides  $b$  if there exists a number  $c$  such that  $ac = b$ .

This is written as  $a|b$ .

We call  $a$  a factor or divisor of  $b$  and we call  $b$  a multiple of  $a$

## Definition: Quotient and remainder

Let  $a$  be an integer and  $d$  a positive integer. Then there exist unique integers  $q$  and  $r$ , with  $0 \leq r < d$ , such that

$$a = dq + r$$

We call  $d$  the **divisor**,  $a$  the **dividend**,  $q = a \text{ div } d$  the **quotient**, and  $r = a \bmod d$  the **remainder**.

# 2 Modular inverse

## Definition: Modular inverse

Modular inverse, is the concept that for any integer  $a$ , The modulo is an integer  $m$ , where there exists  $x$  such that:

$$a \cdot x \equiv 1 \bmod m$$

This exists on the condition that  $a$  and  $m$  are **coprime**

## Definition: Coprime

Two integers  $a, m$  are coprime if and only if

$$\gcd(a, m) = 1$$

## Example: Find the modular inverse

Given the integers  $a = 3$  and  $m = 14$

To calculate the modular inverse, it essentially means that we want to express 1 as a combination of 3 and 14. First we will first calculate their gcd.

$\text{gcd}(3, 14)$

$$14 = 4 \cdot 3 + 2$$

$$3 = 1 \cdot 2 + 1$$

$$2 = 2 \cdot 1 + 0$$

Since the last coefficient before achieving 0 was 1. Their GCD is 1

We then use the Extended Euclidean Algorithm by making our way back from  $3=1 \cdot 2+1$ .

$$1 = 3 - 2 \cdot 1 \quad \text{where } 2 = 14 - 4 \cdot 3$$

$$1 = 3 - 1 \cdot (14 - 4 \cdot 3)$$

$$1 = 3 - 14 + 4 \cdot 3$$

$$1 = 5 \cdot 3 - 14$$

This means that

$$1 = 5 \cdot 3 - 1 \cdot 14$$

$$x = 5$$

Here's the python 🐍 implementation

| Inverse Mod |                                             | Python |
|-------------|---------------------------------------------|--------|
| 1           | <code>def inverse_mod(a, b):</code>         |        |
| 2           | <code>    x = a</code>                      |        |
| 3           | <code>    y = b</code>                      |        |
| 4           | <code>    oldolds = 1</code>                |        |
| 5           | <code>    olds = 0</code>                   |        |
| 6           | <code>    oldoldt = 0</code>                |        |
| 7           | <code>    oldt = 1</code>                   |        |
| 8           | <code>    while y != 0:</code>              |        |
| 9           | <code>        q = x // y</code>             |        |
| 10          | <code>        r = x % y</code>              |        |
| 11          | <code>        x = y</code>                  |        |
| 12          | <code>        y = r</code>                  |        |
| 13          | <code>        s = oldolds - q * olds</code> |        |
| 14          | <code>        t = oldoldt - q * oldt</code> |        |
| 15          | <code>        oldolds = olds</code>         |        |
| 16          | <code>        oldoldt = oldt</code>         |        |
| 17          | <code>        olds = s</code>               |        |
| 18          | <code>        oldt = t</code>               |        |
| 19          | <code>    return oldolds</code>             |        |

### 3 GCD

#### Definition: GCD

The greatest common divisor is the greatest integer that divides both  $a$  and  $b$ .

This can be found by integers  $a, b$  such that when finding the modular inverse of  $a, b$ , the last nonzero answer will be their GCD

#### Example: GCD

Find  $\gcd(10, 36)$

This is done by definition “Definition: GCD”

$$36 \bmod 10 \equiv 6$$

$$10 \bmod 6 \equiv 4$$

$$6 \bmod 4 \equiv 2$$

$$4 \bmod 2 \equiv 0$$

Thus,  $\gcd(10, 36) = 2$

## 4 LCM

### Definition: LCM

The Least Common Multiple of the positive integers  $a, b$  is the smallest positive integer that is divisible by both  $a$  and  $b$ . The least common multiple of  $a$  and  $b$  is denoted by  $\text{lcm}(a, b)$

Let  $a = p_1^{a_1} \cdot p_2^{a_2} \cdot \dots \cdot p_n^{a_n}$  and  $b = p_1^{b_1} \cdot p_2^{b_2} \cdot \dots \cdot p_n^{b_n}$  be the prime factorizations of  $a$  and  $b$ , where  $p_1, p_2, \dots, p_n$  are the distinct primes dividing  $a$  or  $b$ , and each exponent  $a_i, b_i \geq 0$ .

Then  $\text{lcm}(a, b)$  is defined as

$$\text{lcm}(a, b) = p_1^{\max(a_1, b_1)} \cdot p_2^{\max(a_2, b_2)} \cdot \dots \cdot p_n^{\max(a_n, b_n)}$$

### Definition: Prime factorization

- 1) Start with the smallest prime number ( $p_i$ ) and check if it divides the given number. Repeat until it's no longer divisible by  $p_i$ .
- 2) Move to next prime ( $p_{i+1}$ ), and repeat with the remainder of step 1
- 3) Express the result as a product of primes.

### Example: Prime factorization

Find the prime factorization of 60.

$$\frac{60}{2} = 30, \quad \frac{30}{2} = 15$$

15 is not divisible by 2, so we divided by 2 twice and continue with 15.

$$\frac{15}{3} = 5$$

We divided by 3 once. 5 is already prime, so it counts once.

Thus, 60 can be written as  $60 = 2^2 \cdot 3 \cdot 5$ .

## 5 Euclidean algorithm

### Definition: Euclidean algorithm

The Euclidean algorithm is a method to find the greatest common divisor between two numbers.

Let  $a = b \cdot q + r$

Where  $a$  is the greater of the two numbers,  $b$  is the smaller of the two numbers,  $q$  is the quotient (how many times  $b$  divides  $a$ ,  $\lfloor a/b \rfloor = q$ ), and  $r$  is the remainder of  $a \bmod b$

Repeat this sentence with the remainder, and stop when  $r = 0$ , the GCD will be the last nonzero remainder

Written recursively in python 🐍 here

### Greatest Common Divider

Python

```
1 def gcd(a, b):  
2     if b == 0:  
3         return a  
4     return gcd(b, a % b)
```

### Example: Euclidean algorithm

Find  $\gcd(18, 12)$ .

$$18 = 12 \cdot 1 + 6$$

$$12 = 6 \cdot 2 + 0$$

Since the remainder is now 0, the GCD is the last nonzero remainder.

Thus,  $\gcd(18, 12) = 6$ .

$$\phi(x_i) = e^{-\alpha(c_j^2 - r_j^2)}$$

## 6 Chinese Remainder Theorem

### Definition: The Chinese Remainder Theorem

The Chinese Remainder Theorem is a method used to solve simultaneous congruences with different moduli.

Consider a system of simultaneous congruences

$$\begin{aligned}x &\equiv a_1 \pmod{n_1} \\x &\equiv a_2 \pmod{n_2} \\&\vdots \\x &\equiv a_k \pmod{n_k}\end{aligned}$$

Requirements for the Chinese Remainder Theorem:

- $n_1, n_2, \dots, n_k$  are pairwise coprime, i.e.  $\gcd(n_i, n_j) = 1$  for all  $i \neq j$ .
- $a_1, a_2, \dots, a_k$  are known integers.

When these requirements are met, the Chinese Remainder Theorem guarantees there is a unique solution for  $x \bmod N$ , where

$$N = n_1 \cdot n_2 \cdot \dots \cdot n_k$$

### Example: Chinese Remainder Theorem

Solve the following system of congruences:

$$\begin{aligned}x &\equiv 2 \pmod{3} \\x &\equiv 3 \pmod{5} \\x &\equiv 1 \pmod{7}\end{aligned}$$

1) **Compute  $N$** , the product of the moduli.

$$N = 3 \cdot 5 \cdot 7 = 105$$

2) **Compute the partial products  $N_i = N/n_i$ .**

$$N_1 = \frac{105}{3} = 35, \quad N_2 = \frac{105}{5} = 21, \quad N_3 = \frac{105}{7} = 15$$

3) **Find the modular inverses  $y_i$**  such that  $N_i \cdot y_i \equiv 1 \pmod{n_i}$ .

- $N_1 \equiv 35 \equiv 2 \pmod{3}$ , and  $2 \cdot y_1 \equiv 1 \pmod{3}$  gives  $y_1 = 2$ .
- $N_2 \equiv 21 \equiv 1 \pmod{5}$ , and  $1 \cdot y_2 \equiv 1 \pmod{5}$  gives  $y_2 = 1$ .
- $N_3 \equiv 15 \equiv 1 \pmod{7}$ , and  $1 \cdot y_3 \equiv 1 \pmod{7}$  gives  $y_3 = 1$ .

4) **Construct the solution  $x = \sum_i a_i \cdot N_i \cdot y_i$ .**



$$\begin{aligned}
 x &= (2 \cdot 35 \cdot 2) + (3 \cdot 21 \cdot 1) + (1 \cdot 15 \cdot 1) \\
 &= 140 + 63 + 15 \\
 &= 218
 \end{aligned}$$

5) **Reduce  $x$  modulo  $N$ .**

$$218 \bmod 105 = 8$$

Thus, the solution is  $x \equiv 8 \pmod{105}$ .

Chinese remainder theorem implemented in python 🐍

Chinese Remainder Theorem Python

```

1  def chi_rem_thm(mn,an):
2      m = 1
3      Mn = []
4      yn = []
5      for k in range(0, len(mn)):
6          m = m * mn[k]
7      for k in range (0, len(mn)):
8          Mk = m / mn[k]
9          Mn.append(Mk)
10         yk = inverse_mod(Mn[k],mn[k]) % mn[k]
11         yn.append(yk)
12     x = 0
13     for k in range (0, len(yn)):
14         x = x + an[k] * Mn[k] * yn[k]
15     while x >= m:
16         x = x - m
17     return x

```